

FAULT CODE 141 - Engine Oil Pressure/Temperature Sensor Circuit

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead, and Part Number 3162898 - breakout cable.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for sensor supply fault codes.	Sensor supply fault codes active?
	STEP 1B. Check for an active fault code.	Fault Code 141 active?
STEP 2.	Check the oil pressure/temperature sensor and circuit.	
	STEP 2A. Inspect the oil pressure/temperature sensor and connector pins.	Dirty or damaged pins?
	STEP 2B. Check the sensor supply voltage and return circuit.	4.75 VDC to 5.25 VDC?
STEP 3.	Check the ECM and engine harness.	

STEPS		SPECIFICATIONS
	STEP 3A. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check the sensor supply voltage and return circuit.	4.75 VDC to 5.25 VDC?
	STEP 3C. Inspect the ECM and engine harness connector pins.	Dirty or damaged pins?
	STEP 3D. Check for an open circuit in the original equipment manufacturer (OEM) harness signal side.	Less than 10 ohms?
	STEP 3E. Check for an open circuit in the OEM harness return side.	Less than 10 ohms?
	STEP 3F. Check for a short circuit to ground in the OEM harness.	Greater than 100k ohms?
	STEP 3G. Check for a pin-to-pin short circuit in the OEM harness.	Greater than 100k ohms?
	STEP 3H. Check for an active fault code.	Fault Code 141 active?
STEP 4.	Check ECM calibration and clear fault codes.	
	STEP 4A. Check if an ECM calibration update is available.	If a calibration update for this fault code is available, does the ECM contain that revision or higher?
	STEP 4B. Disable the fault code.	Fault code active?

STEP 1. Check the fault codes.

STEP 1A. Check for sensor supply fault codes.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for sensor supply fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Sensor supply fault codes active? YES	Appropriate fault code troubleshooting tree
	Sensor supply fault codes active? NO	1B

STEP 1B. Check for an active fault code.**Conditions :**

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 141 active? YES	2A
	Fault Code 141 active? NO	Use the following procedure for an inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)

STEP 2. Check the oil pressure/temperature sensor and circuit.**STEP 2A. Inspect the oil pressure/temperature sensor and connector pins.****Conditions :**

- Turn keyswitch OFF.
- Disconnect the oil pressure/temperature sensor from the engine harness.

Action	Specification/Repair	Next Step
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Inspect the engine harness and oil pressure/temperature sensor connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the sensor or harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	2B

STEP 2B. Check the sensor supply voltage and return circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the oil pressure/temperature sensor from the engine harness.
- Install the breakout cable, Part Number 3162898, between the sensor and the engine harness connector.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
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Check the supply voltage and return circuit. Measure the voltage from the oil pressure +5 volt SUPPLY pin to the oil pressure RETURN pin at the sensor connector of the engine harness. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general multimeter usage techniques. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	4.75 VDC to 5.25 VDC? YES	3C
	4.75 VDC to 5.25 VDC? NO	3A

STEP 3. Check the ECM and engine harness.

STEP 3A. Inspect the ECM and engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness from the ECM.

Action	Specification/Repair	Next Step

Inspect the engine harness and ECM engine connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM engine connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	3B

STEP 3B. Check the sensor supply voltage and return circuit.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness from the ECM. • Install the breakout cable, Part Number 3162898, between the sensor and the engine harness connector. • Turn keyswitch ON.		
Action	Specification/Repair	Next Step

Check the supply voltage and return circuit. Measure the voltage from the oil pressure +5 volt SUPPLY pin to the oil pressure RETURN pin at the ECM. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general multimeter usage techniques. Refer to Procedure 019-359 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)	4.75 VDC to 5.25 VDC? YES	3C
	4.75 VDC to 5.25 VDC? NO Repair : Replace the ECM. Refer to Procedure 019-031 in the Associated Procedures Table.	4A

STEP 3C. Inspect the ECM and engine harness connector pins.

Conditions : - Turn keyswitch OFF. - Disconnect the engine harness from the ECM.		
Action	Specification/Repair	Next Step

Inspect the engine harness and ECM engine connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? YES Repair : A damaged connection has been detected in the ECM engine connector or engine harness connector. Clean the connector and pins. Repair the damaged harness, connector, or pins, if possible. Refer to Procedure 019-043 in the Associated Procedures Table.	4A
	Dirty or damaged pins? NO	3D

STEP 3D. Check for an open circuit in the OEM harness signal side.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM.
- Disconnect the oil pressure/temperature sensor from the OEM harness.

Action	Specification/Repair	Next Step
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Check for an open circuit. Measure the resistance between the OEM harness ECM connector oil pressure/temperature sensor SIGNAL pin and the OEM harness oil pressure/temperature sensor connector SIGNAL pin. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3E
	Less than 10 ohms? NO Repair : An open signal circuit has been detected in the OEM harness. Troubleshoot each harness connected in series to determine which contains the open signal circuit. Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3E. Check for an open circuit in the OEM harness return side.

Conditions : - Turn keyswitch OFF. - Disconnect the OEM harness from the ECM. - Disconnect the oil pressure/temperature sensor from the OEM harness.		
Action	Specification/Repair	Next Step

Check for an open circuit. • Measure the resistance between the OEM harness ECM connector oil pressure/temperature sensor RETURN pin and the OEM harness oil pressure/temperature sensor connector RETURN pin. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	3F
	Less than 10 ohms? NO Repair : An open return circuit has been detected in the OEM harness. Troubleshoot each harness connected in series to determine which contains the open return circuit. Repair or replace the OEM harness. • Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3F. Check for a short circuit to ground in the OEM harness.

Conditions : • Turn keyswitch OFF. • Disconnect the OEM harness from the ECM. • Disconnect the oil pressure/temperature sensor from the OEM harness.		
Action	Specification/Repair	Next Step

Check for a short circuit to ground. Measure the resistance between the oil pressure/temperature sensor SIGNAL pin in the OEM harness ECM connector and chassis ground. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3G
	Greater than 100k ohms? NO Repair : A short circuit to ground on the SIGNAL wire has been detected in the OEM harness. Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3G. Check for a pin-to-pin short circuit in the OEM harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness from the ECM.
- Disconnect the oil pressure/temperature sensor from the OEM harness.

Action	Specification/Repair	Next Step

Check for a pin-to-pin short circuit. Measure the resistance between the oil pressure/temperature sensor SIGNAL pin in the OEM harness ECM connector and all other pins in the OEM harness ECM connector. Refer to the circuit diagram or the wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Greater than 100k ohms? YES	3H
	Greater than 100k ohms? NO Repair : A pin-to-pin short circuit on the SIGNAL wire has been detected in the OEM harness. Repair or replace the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	4A

STEP 3H. Check for active fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE electronic service tool.

Action	Specification/Repair	Next Step
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Check for the appropriate circuit response after 30 seconds. • Use INSITE electronic service tool to read the fault codes.	Fault Code 141 active? YES Repair : A damaged sensor has been detected. Replace the oil pressure/temperature sensor. Refer to Procedure 019-155 in the Associated Procedures Table.	4A
	Fault Code 141 active? NO Repair : None. The removal and installation of the connector corrected the fault.	4A

STEP 4. Check ECM calibration and clear fault codes.

STEP 4A. Check if an ECM calibration update is available.

Conditions :

- Connect all components.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Compare the ECM code and revision number in the ECM to the calibration revisions listed in the ECM Calibration Revision History for applicable changes related to this fault code. Use INSITE™ electronic service tool to find the present ECM code and revision number in the ECM. The ECM code and revision number are found in the Calibration Information section of System ID and Dataplate in Features and Parameters.	If a calibration update for this fault code is available, does the ECM contain that revision or higher? YES	4B
	If a calibration update for this fault code is available, does the ECM contain that revision or higher? NO Repair : If necessary, calibrate the ECM. Refer to Procedure 019-032 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)	4B

STEP 4B. Disable the fault code.

Conditions : - Connect all components. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Disable and clear the fault code. Operate the engine within the "Conditions for Clearing the Fault Code" found in the Overview section of this troubleshooting procedure.	Fault code active? YES Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A
	Fault code active? NO	Repair complete

Associated Procedures

Associated Procedures			
Procedure Title	Procedure Number	Service Model Name	Bulletin Number
Engine Control Module	Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html)	ISM11 CM570 QSM11 CM570	3666266
Engine Wiring Harness	Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/82/82-019-043-tr.html)	ISM11 CM570 QSM11 CM570	3666266
Oil Pressure/Temperature Sensor	Refer to Procedure 019-155 (/qs3/pubsys2/xml/en/procedures/82/82-019-155.html)	ISM11 CM570 QSM11 CM570	3666266